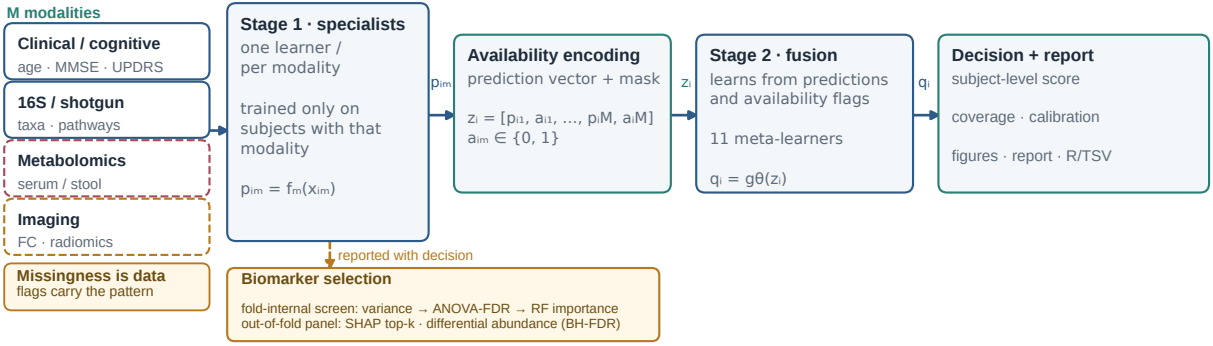
AVAIL: availability-aware multimodal integration

Every subject contributes; absence is represented, not discarded or fabricated.

(a) Availability-aware pipeline

one canonical path; no modality-specific workflows



(b) Validation envelope

the same pipeline is exercised end to end; no post-hoc selection

Leak-free evaluation
5-fold / LOO
fold-internal preprocessing
no test leakage

Leakage control
full-pipeline label
permutation null
empirical null

Missingness stress
MCAR / MAR / MNAR
block dropout
coverage strata

External cohorts
MetaCardis · HMP2
ds004392
coverage audit